

Scalability & Future Plan

Date	4 July 2026
Team ID	SWTID-2026-5853
Project Name	A Comprehensive Measure of Well-Being (Human Development Index Prediction)
Maximum Marks	3 Marks

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Limited concurrent user support	Performance may decrease under heavy load	High
2	SQLite database for current deployment	Not ideal for very large datasets	High
3	Manual dataset updates	Latest HDI values require manual refresh	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask server	Deploy with Gunicorn, Nginx, and cloud auto-scaling
2	Data Storage	SQLite	Migrate to PostgreSQL/MySQL
3	Performance	Basic optimization	Caching, optimized queries, background jobs
4	Security	Login authentication	JWT, HTTPS, role-based access control, audit logs

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy application to cloud platform	1 Month	Improved availability
Phase 2	Real-time HDI dataset integration	2 Months	More accurate predictions
Phase 3	Interactive analytics dashboard and world map	3 Months	Better insights
Phase 4	Mobile application and multilingual support	6 Months	Wider adoption